Department of Computer Science and Engineering (Data Science)
B.Tech. Sem: III Subject: Statistics for Data Science
Experiment 5

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Date:	**Experiment Title: Confidence Interval**
Aim	To implement Confidence Interval using Python.
Software	Google Colab
Implementation	Using Python solve the following questions given below :

1. *If Z follows standard normal distribution, then find*

$(i)\ P(Z<1.5)$ $(ii)\ P(Z>0.5)$ $(iii)\ P(|Z|<1.5)$ $(iv)\ P(|Z|>0.5)$

$(v)\ P(-2.2<Z<1)$

Code:

```
[2] from scipy.stats import norm
    norm.cdf(1.5)
```

```
0.9331927987311419
```

```
[ ] 1-norm.cdf(0.5)
```

```
0.3085375387259869
```

```
(norm.cdf(1.5)-0.5)*2
```

```
0.8663855974622838
```

```
[ ] 1-(norm.cdf(0.5)-0.5)*2
```

```
0.6170750774519738
```

```
[ ] norm.cdf(2.2)-norm.cdf(1)
```

```
0.14475180641795848
```

2. *If Z follows standard normal distribution, then find value of* Z_0 *satisfying the given equation*

$(i)\ P(Z<Z_0)=0.90$ $(ii)\ P(Z<Z_0)=0.95$ $(iii)\ P(Z<Z_0)=0.99$

$(iv)\ P(|Z|<Z_0)=0.90$ $(v)\ P(|Z|<Z_0)=0.95$ $(vi)\ P(|Z|<Z_0)=0.99$

Code:

```
[ ] norm.ppf(0.9)
```

1.2815515655446004

```
[ ] norm.ppf(0.95)
```

1.6448536269514722

```
norm.ppf(0.99)
```

2.3263478740408408

```
[ ] norm.ppf(0.95)
```

1.6448536269514722

```
[ ] norm.ppf(0.975)
```

1.959963984540054

```
[ ] norm.ppf(0.995)
```

2.5758293035489004

3. $If\ t\ follows\ students\ t\ distribution, then\ find$

$(i)\ P(t < 1.5)\ with\ d.o.f. = 20$ $(ii)\ P(t > 0.5)\ with\ d.o.f. = 15$

$(iii)\ P(|t| < 1.5)\ with\ d.o.f. = 25$ $(iv)\ P(|t| > 0.5)\ with\ d.o.f. = 35$

$(v)\ P(-2.2 < t < 1)\ with\ d.o.f. = 42$

Code:

```
[43] from scipy.stats import t
     t.cdf(1.5,20)

0.9253821144153737
```

```
1-t.cdf(0.5,15)

0.3121650567600378
```

```
[ ] t.cdf(1.5,25)-t.cdf(-1.5,25)

0.8538615348619807
```

```
[ ] 1-(t.cdf(0.5,25)-t.cdf(-0.5,25))

0.6214477851902287
```

```
[ ] t.cdf(1,42)-t.cdf(-2.2,42)

0.8218007171327162
```

4. *If t follows students t distribution, then find value of* t_0 *satisfying the given equation*

(i) $P(t < t_0) = 0.90$ *with* $d.o.f. = 20$ (ii) $P(t < t_0) = 0.95$ *with* $d.o.f. = 15$

(iii) $P(t < t_0) = 0.99$ *with* $d.o.f. = 25$ (iv) $P(|t| < t_0) = 0.90$ *with* $d.o.f. = 30$

(v) $P(|t| < t_0) = 0.95$ *with* $d.o.f. = 42$ (vi) $P(|t| < t_0) = 0.99$ *with* $d.o.f. = 10$

Code:

```
[29] t.ppf(0.9,20)
1.3253407069850462

t.ppf(0.95,15)
1.7530503556925547

t.ppf(0.99,25)
2.4851071754106413

[ ] t.ppf(0.95,30)
1.6972608943617378

[ ] t.ppf(0.975,42)
2.018081697095881

[ ] t.ppf(0.995,10)
3.169272667175838
```

5. *If F follows Snedecor's F distribution, then find*

$(i)\ P(F < 1.5)\ with\ df_1 = 5, df_2 = 14$ $\quad(ii)\ P(F > 2.5)\ with\ df_1 = 15, df_2 = 14$

$(iii)\ P(0.5 < F < 4.1)\ with\ df_1 = 13, df_2 = 17$

Code:

```
from scipy.stats import f
f.cdf(1.5,5,14)
```

```
0.7480582329156877
```

```
1-f.cdf(2.5,15,14)
```

```
0.04734368894060448
```

```
f.cdf(4.1,13,17)-f.cdf(0.5,13,17)
```

```
0.8911203850553827
```

6. *If F follows Snedecor's F distribution, then find value of F_0 satisfying the given equation*

(i) $P(F < F_0) = 0.90$ *with* $df_1 = 5, df_2 = 14$

(ii) $P(F < F_0) = 0.95$ *with* $df_1 = 15, df_2 = 13$

(iii) $P(F < F_0) = 0.99$ *with* $df_1 = 25, df_2 = 28$

```
f.ppf(0.9,5,14)
```

```
2.3069430514007236
```

```
f.ppf(0.95,15,13)
```

```
2.533109983130745
```

```
f.ppf(0.99,25,28)
```

```
2.5060172667359417
```

Code:

7. $If\ X\ follows\ \chi^2 - distribution, then\ find$

$(i)\ P(X < 1.5)\ with\ df = 10$ $(ii)\ P(X > 2.5)\ with\ df = 5$

$(iii)\ P(0.5 < X < 4.1)\ with\ df = 2$

Code:

```
[ ] from scipy.stats import chi2
    chi2.cdf(1.5,10)

    0.0010646777727857928
```

```
[ ] 1-chi2.cdf(2.5,5)

    0.7764950711233227
```

```
[ ] chi2.cdf(4.1,2)-chi2.cdf(0.5,2)

    0.6500658794836006
```

8. $If\ X\ follows\ \chi^2 - distribution, then\ find\ value\ of\ X_0$ $satisfying\ the\ given\ equation$

$(i)\ P(X < X_0) = 0.90\ with\ df = 1$ $(ii)\ P(X < X_0) = 0.95\ with\ df = 3$

$(iii)\ P(X < X_0) = 0.99\ with\ df = 2$ $(iv)\ P(X < X_0) = 0.05\ with\ df = 1$

$(v)\ P(X < X_0) = 0.025\ with\ df = 3$ $(vi)\ P(X < X_0) = 0.005\ with\ df = 2$

Code:

```
[ ] chi2.ppf(0.9,1)
```

2.705543454095404

```
[ ] chi2.ppf(0.95,3)
```

7.814727903251179

```
[ ] chi2.ppf(0.99,2)
```

9.21034037197618

```
chi2.ppf(0.05,1)
```

0.003932140000019522

```
[ ] chi2.ppf(0.025,3)
```

0.21579528262389785

9. *Construct a* 95 % *confidence interval for population mean in an experiment that found the sample mean temperature for a certain city in August was* 101.82, *with a population standard deviation of* 1.2.*There were* 6 *samples in this experiment*.

Code:

```
import math
import numpy as np
z=norm.ppf(0.95)
l1=101.82-(z*12)/np.sqrt(6)
print("L=",l1)
l2=101.82+(z*12)/np.sqrt(6)
print("U=",l2)
print("Confidence interval is",l1,",",l2)
```

```
L= 93.76189582480532
U= 109.87810417519466
Confidence interval is 93.76189582480532 , 109.87810417519466
```

	10. *Construct a* 98% *Confidence Interval for population mean based on the following data:*
	
	45, 55, 67, 45, 68, 79, 98, 87, 84, 82.
	
	Code:
	`[32] l=[45,55,67,45,68,79,98,87,84,82]`
	`k=np.mean(l)`
	`std=np.std(l)`
	
	`[46] t=t.ppf(0.99,9)`
	
	`L=k-(t*std/np.sqrt(9))`
	`U=k+(t*std/np.sqrt(9))`
	`print("Confidence interval is")`
	`print(L,",",U)`
	
	`Confidence interval is`
	`54.7866168002499 , 87.2133831997501`
	
	11. 510 *people applied to the Bachelor's in Elementary Education program at Florida State College. Of those applicants,* 57 *were men. Find the* 90% *CI of the true proportion of men who applied to the program.*
	
	Code:
	`n=510`
	`sp=57/510`
	`z=norm.ppf(0.95)`
	`k=np.sqrt(sp*(1-sp)/n)`
	`L=sp-z*k`
	`U=sp+z*k`
	`print("Confidence interval is",L,",",U)`
	
	`Confidence interval is 0.08881598300884237 , 0.13471342875586353`

Conclusion	Confidence interval has been successfully implemented using python libraries.

Signature of Faculty